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ORIGINAL PAPER

Enriching the Korean learner corpus for grammatical error correction and writing assessment

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Abstract

Despite growing global interest in Korean language education, learner corpora tailored to Korean L2 writing remain scarce. This paper introduces KoLLA v2.0, the first Korean learner corpus to incorporate both multi-reference GEC annotations and rubric-based essay scoring. We extend the original KoLLA dataset by adding a second human correction for each sentence, creating the first multi-reference GEC resource for Korean. This design captures the variability of valid corrections in an agglutinative and morphologically complex language, enabling fairer and more realistic evaluation of GEC systems. In parallel, we enrich the corpus with rubric-based scores based on criteria from the Korean National Language Institute, providing standardized, multi-dimensional assessments of grammatical accuracy, coherence, and lexical diversity. These enhancements position KoLLA v2.0 as a standardized resource for research in Korean L2 learning and instruction, and as a benchmark for evaluating automated error correction and essay scoring systems.

Keywords Korean learner corpus · Grammatical error correction · Multi-reference GEC · Rubric-based scoring · KoLLA corpus

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1 Introduction

Written and spoken data produced by second language (L2) learners have long served as valuable resources for studying L2 acquisition. With the rise of electronic learner corpora, researchers now benefit from systematic and scalable analyses of learner errors, interlanguage patterns, and developmental stages (Zhang and Fu, 2024). Tools such as the UCLouvain Error Editor and ERRANT have further standardized error annotation, supporting reproducible and cross-linguistic studies (Bryant et al., 2017; Bryant, 2019).

Despite these advances for widely studied languages such as English and Chinese, Korean remains underrepresented in learner corpus research. This gap is striking given the rapid global expansion of Korean language education, driven by cultural interest and a growing population of Korean L2 learners. According to the Modern Language Association, Korean ranked among the top ten most-taught languages in U.S. higher education between 2013 and 2016.[1] Yet existing Korean learner corpora, such as KoLLA and the KyungHee corpus, remain limited in scope: they lack multi-reference grammatical error correction data and rubric-based scoring, both of which are central to current grammatical error correction (GEC) and automated essay scoring (AES) research.

The present work introduces **KoLLA v2.0** as a response to the limited error-annotation depth and the absence of rubric-based scoring in existing Korean learner corpora. Building on the original KoLLA dataset (Lee et al., 2012), we contribute two novel enhancements. First, we add a second human correction for each learner sentence, making KoLLA v2.0 the first multi-reference GEC corpus for Korean. This design captures the variability of valid corrections in an agglutinative and morphologically complex language, enabling fairer and more realistic evaluation of GEC systems. Second, we enrich the corpus with rubric-based scores aligned with the Korean National Language Institute's guidelines, highlighting their capacity to assess higher-order writing features such as discourse coherence and lexical diversity–dimensions that traditional, surface-focused GEC methods often fail to capture. This addition provides a more nuanced and pedagogically relevant framework for evaluating learner writing, supporting both comparative analysis and instructional design.

In this expanded form, KoLLA v2.0 serves a dual role: it is both a benchmark resource for developing and evaluating NLP systems for Korean and a pedagogically meaningful dataset for studying L2 learning and informing classroom practice. The corpus comprises 100 essays written by beginner- and intermediate-level Korean L2 learners (heritage and non-heritage), segmented into 1,418 sentences with two reference corrections per sentence (2,836 references in total). Each essay is further evaluated with rubric scores assigned by trained annotators. This enriched resource establishes a standardized dataset for Korean learner language, supporting advances in both computational modeling and language pedagogy.

[1] https://www.mla.org/content/download/191324/file/Enrollments-in-Languages-Other-Than-English-in-US-Institutions-of-Higher-Education-Fall-2021.pdf?utm_source=chatgpt.com.

2 Previous works

GEC and AES have been extensively studied for widely taught languages such as English and Chinese, leading to major advances in learner corpus development, error annotation, and scoring frameworks (Bryant et al., 2017; Lin, 2024; Gu et al., 2025; Ridley et al., 2021; He et al., 2022). Large-scale annotated corpora in these languages have provided the foundation for both empirical research and computational applications. In contrast, research and resources for Korean remain limited, with only a handful of learner corpora and studies addressing Korean GEC or AES.

In the context of GEC, annotated corpora have been critical in enabling detailed error analyses and supporting the development of correction systems. For example, the First Certificate in English (FCE) corpus, a subset of the Cambridge Learner Corpus (CLC) (Yannakoudakis et al., 2011), and the multi-reference MuCGEC corpus for Chinese (Zhang et al., 2022) have each supported robust evaluation of GEC models. These corpora typically include aligned learner sentences with one or more corrections, allowing researchers to analyze error patterns across proficiency levels and to develop targeted instructional strategies.

In the context of AES, datasets such as the Automated Student Assessment Prize (ASAP) collection have been foundational. ASAP, which is limited to English, provides rubric-based essay scores across multiple prompts, enabling models to assess essay quality with respect to coherence, fluency, and vocabulary. The enhanced ASAP++ dataset, also English-only, adds attribute-specific scores for sentence structure, word choice, and organization (Mathias and Bhattacharyya, 2018), facilitating finer-grained evaluation. These developments underscore the contrast with the situation for Korean. For Korean, however, relatively few learner corpora exist. The KoLLA corpus, developed by Indiana University and Wellesley College, represents one of the earliest efforts, focusing on postpositional particle errors in essays written by beginner and intermediate learners (heritage and non-heritage). It contains approximately 12,000 words and 1419 sentences from 100 essays (Lee et al., 2012). The Kyunghee corpus (Park and Lee, 2016) is larger in scale, with 3,700 writing samples across multiple proficiency levels, and has been used for tasks such as POS tagging and syntactic parsing. Yet both resources are constrained: KoLLA emphasizes functional morpheme errors without broader multi-reference correction, while Kyunghee lacks detailed error annotation and provides only holistic scores, limiting its usefulness for GEC and AES research.

Beyond corpus development, early work on Korean GEC focused on detecting particle errors using statistical approaches (Israel et al., 2013), while more recent studies have shifted to neural architectures. Copy-augmented transformers (Zhao et al., 2019), originally applied to English GEC, have been adapted for Korean (Min et al., 2020; Lee et al., 2021; Seonmin et al., 2021). More recently, BART (Lewis et al., 2020) has been shown to provide strong baselines for Korean GEC as well (Yoon et al., 2023).

Against this background, this study addresses the limited availability of Korean learner corpora with multi-reference GEC and rubric-based scoring, by presenting KoLLA v2.0, the first Korean learner corpus to incorporate (i) multi-reference GEC annotations, which capture the variability of valid corrections in a morphologically

complex language, and (ii) rubric-based essay scoring aligned with Korean National Language Institute guidelines. These additions establish KoLLA v2.0 as the first standardized resource for Korean that supports both GEC and AES research, bridging the gap between existing learner corpora and the growing demand for advanced Korean L2 resources.

3 Development and annotation

This section presents two major enhancements to the KoLLA learner corpus: a multi-reference GEC dataset and a rubric-based AES dataset. These additions directly address gaps in existing Korean learner resources. The multi-reference GEC dataset provides two corrected versions for each sentence, capturing linguistic variability and enabling more faithful evaluation of GEC systems. The rubric-based AES dataset, aligned with the Korean National Language Institute's guidelines, moves beyond holistic scoring to offer detailed rubric-based evaluations, thereby improving the reliability and pedagogical relevance of L2 writing assessment. In combination, these contributions establish KoLLA v2.0 as the first Korean learner corpus to integrate both multi-reference GEC annotations and rubric-based scoring, offering a standardized resource for advancing Korean L2 acquisition research, instructional design, and automated assessment.

3.1 Introduction to KoLLA

The KoLLA learner corpus addresses the challenge of annotating and analyzing Korean post-positional particle errors, a persistent difficulty for learners of Korean as a L2 (Lee et al., 2012). These particles, comparable to English prepositions, fulfill multiple functions, including marking grammatical roles (e.g., subject and object), semantic roles (e.g., agent and location), and discourse functions. The agglutinative nature of Korean and the tendency for particles to attach to nominals without whitespace further complicate their acquisition.

The original KoLLA corpus consists of essays written by college-level Korean learners, annotated specifically for postpositional particle errors. Its annotation scheme is multi-layered: spacing and spelling errors are first corrected to ensure proper segmentation, after which particle errors are identified and classified. Particle errors are divided into four categories: omission, substitution, addition, and ordering.

To determine correct particle usage, the corpus follows the principle of "minimal interaction," requiring only that the corrected form be grammatical with respect to particle use in formal Korean writing. In cases where multiple particles are equally valid, the annotation allows sets of possible corrections, ensuring more robust evaluation of automatic systems.

Developed from 100 learner essays and encoded using EXMARaLDA (Schmidt, 2010), the original KoLLA corpus achieved high inter-annotator agreement, validating both its guidelines and methodology. By supporting automatic particle error detection, it provided a valuable resource for research on morphologically rich languages and contributed best-practice insights for learner corpus annotation.

The original KoLLA dataset is composed of four learner subgroups: foreign beginners (FB), foreign intermediates (FI), heritage beginners (HB), and heritage intermediates (HI). Each subgroup consists of 25 essays, yielding a total of 100 learner essays collected from American university students.

The division reflects both background and proficiency. Foreign learners are those who began Korean exclusively in classroom contexts, without prior exposure in the home environment. Heritage learners, by contrast, are students who had early exposure to Korean in family settings, such as through parents or relatives. Within each background, learners were further separated into beginner and intermediate levels, corresponding to their university course placement at the time of data collection.

This design allows for a balanced comparison across both background (heritage vs. foreign) and proficiency (beginner vs. intermediate), thereby enabling analyses of how particle errors vary according to learners' linguistic experience and developmental stage.

Building on this foundation, the present study uses the original resource directly. The 100 essays were segmented into 1,419 sentences to align with the evaluation levels required for both AES and GEC tasks: AES is conducted at the essay level, while GEC typically requires sentence-level annotation.

To enable multi-reference evaluation for GEC, each sentence now has two human-generated corrections. The first correction derives from the original KoLLA annotations, while the second was newly created by a trained linguist following detailed annotation guidelines. This dual-reference design captures a broader range of valid corrections, which is particularly important for a morphologically rich language such as Korean. For AES, two annotators independently assigned rubric-based scores to all essays, evaluating content, coherence, grammatical accuracy, and vocabulary use.

With these extensions, the present version represents a substantial enrichment of KoLLA rather than an entirely new dataset. While the learner-produced texts remain unchanged, KoLLA v2.0 introduces both multi-reference GEC annotations and rubric-based scoring, following best practices in learner corpus construction and evaluation. The enriched resource (KoLLA v2.0) supports more comprehensive investigations of grammatical error correction and essay scoring in Korean as a foreign language.

3.2 Multi-reference GEC corpus

The original KoLLA corpus, consisting of open writing assignments from Korean L2 learners, serves as the foundation for our enhanced multi-reference GEC dataset (KoLLA v2.0). The corpus includes 1,419 sentences written by beginner- and intermediate-level learners, segmented from essays to facilitate sentence-level annotation, given the difficulty of automatic boundary detection in the presence of learner errors.

To extend this resource, we augmented the original KoLLA corrections with a second human-generated correction for each sentence. This design follows precedents in English and Chinese GEC corpora (e.g., FCE, MuCGEC), where multiple references are employed to capture the inherent ambiguity of learner errors. For Korean, such variability is even more pronounced due to its morphosyntactic flexibility, honorific system, and case-marking ambiguity. In an agglutinative language, a single correc-

tion may fail to represent the full range of valid alternatives, potentially penalizing correct but divergent outputs.

The second layer of annotation was produced by a trained linguist with an M.A. in Korean Language Education, following detailed guidelines to ensure fluency, preserve the learner's intended meaning, maintain structural fidelity, and avoid unnecessary rewriting. Although this second set of corrections was provided by a single annotator, quality assurance emphasized internal consistency through strict adherence to the guidelines. A multi-annotator extension is planned for future work.

As a result of this procedure, the dataset provides two reference corrections per sentence, offering a more realistic and flexible evaluation framework for Korean GEC. By capturing correction variability, it supports fairer and more informative evaluation of system performance.

Grammatical error typology further structures the annotation. Whereas the original KoLLA employed morphologically segmented annotations, KoLLA v2.0 adopts word-level (primarily *eojeol*-based) error labels without morphological tags, reflecting current neural GEC practices–particularly sequence-to-sequence models–which operate directly on surface forms (Qiu et al., 2025). This choice aligns with the widely adopted `errant` framework (Bryant et al., 2017), supporting standardization and cross-linguistic comparability. Building on the MRU framework (Missing, Replacement, Unnecessary), we extend it for Korean with language-specific categories, formalized in Algorithm 1.

Key extensions include:

- **Word boundary errors**, frequent in agglutinative languages, with two categories: `M:WB` (missing boundaries, when tokens are fused) and `U:WB` (unnecessary boundaries, when valid tokens are split).
- **Spelling errors**, labeled `R:SPELL`, covering orthographic or phonological mistakes in content words.
- **Functional morpheme errors**, especially in postpositions (`ADP`) and verbal endings (`PART`). If the content word is correct, errors are annotated by context:

  Missing postposition: $\texttt{R:POS}_i \rightarrow \texttt{POS}_j\texttt{+ADP}$
  Unnecessary postposition: $\texttt{R:POS}_i\texttt{+ADP} \rightarrow \texttt{POS}_j$
  Substitution: $\texttt{R:POS+ADP}_i \rightarrow \texttt{POS+ADP}_j$

If the content word itself is incorrect, the error is subsumed under `R:SPELL`.

- **Word order errors**, marked as `R:WO` when the same lexical items occur but in different sequences.

To reduce additional sources of inconsistency, preprocessing normalized frequent variants in particle use and compound verb boundaries, lowering segmentation misalignment errors to under 5%.

```
S 0비행기 1음식이 2안 3막였습니다 4.
A 1 2|||R:NOUN+ADP → NOUN+ADP|||음식을|||REQUIRED|||-NONE-|||0
A 3 4|||R:SPELL|||먹었습니다|||REQUIRED|||-NONE-|||0
A 3 4|||R:SPELL|||맞았습니다|||REQUIRED|||-NONE-|||1
```

Fig. 1 Examples from the Korean M2 file: Annotator 0 corrected the sentence as *I didn't eat the airplane food*, while Annotator 1 corrected it as *The airplane food didn't agree with me*, resulting in divergent annotation outcomes. For illustrative purposes, index numbers are included to refer to word positions in the source sentence. These indices are not present in the original M2 files but are added here to facilitate discussion, comparison, and reference

Algorithm 1 formalizes these procedures. Figure 1 illustrates how annotators may diverge in correction choices, underscoring the challenge of distinguishing grammatical errors from alternative but grammatically valid expressions.

Algorithm 1 Error annotation procedure

Given a source and target sentence, perform the following checks:
if Associated content word is correct **then**
  if ADP or PART is present unnecessarily in the source sentence **then**
    $\mathcal{A}$ = R:POS$_i$+ADP → POS$_j$
  else if ADP or PART is missing in the source sentence **then**
    $\mathcal{A}$ = R:POS$_i$ → POS$_j$+ADP
  else if ADP or PART is incorrect (substitution needed) **then**
    $\mathcal{A}$ = R:POS+ADP$_i$ → POS+ADP$_j$
  end if
else if Associated content word is incorrect **then**
  $\mathcal{A}$ = R:SPELL
end if
if Word boundary (WB) error occurs **then**
  if Words $w_i w_j$ in source should be separate in target **then**
    $\mathcal{A}$ = M:WB
  else if Words $w_i \sqcup w_j$ in source should be merged in target **then**
    $\mathcal{A}$ = U:WB
  end if
end if
if Word order error occurs ($w_i \sqcup w_j$ in source vs. $w_j \sqcup w_i$ in target) **then**
  $\mathcal{A}$ = R:WO
end if

These annotation strategies and enhancements make KoLLA v2.0 the first Korean learner corpus with multi-reference GEC. By capturing correction variability and accounting for structural nuances specific to Korean, it provides a linguistically informed foundation for developing and evaluating GEC systems tailored to Korean L2 writing.

3.3 Rubric-based scoring dataset

The relationship between first language (L1) and second language (L2) writing has been extensively examined by L2 writing researchers (Hirose, 2003; Keck, 2006; van Weijen et al., 2019). Studies have shown both differences and similarities between L1 and L2 writing in terms of organizational patterns, overall quality, and grammatical accuracy (Hirose, 2001). For instance, Fathman and Kobayashi (1984) observed that many Japanese learners of English preferred inductive (specific-to-general) organi-

zational patterns, contrasting with the deductive (general-to-specific) style favored by American L1 students. However, more advanced Japanese ESL learners tended to adopt deductive organization, suggesting that proficiency and exposure to L2 conventions influence the adoption of writing strategies common in the target language. Similarly, Hirose (2003) found that Japanese EFL students frequently employed deductive structures in both L1 and L2 writing, challenging assumptions that Japanese writers inherently prefer inductive organization. Other research has argued that not only the quality of writing but also the underlying writing processes share important similarities across L1 and L2 (Davoodifard, 2022), supporting the application of L1 process-oriented models in L2 writing research (Hayes, 2000, 2012).

Given these parallels between L1 and L2 writing processes, this study employs a rubric originally developed for evaluating Korean L1 writing to assess Korean L2 learner essays. The rubric comprises three overarching categories–lexis and grammar, organization, and content–each subdivided into multiple operationalized criteria. Widely used in Korean educational contexts, the rubric was designed and refined by experts in writing assessment to ensure clarity, objectivity, and comprehensiveness.

The first category, *lexis and grammar*, includes three subcategories: grammatical accuracy, appropriateness of vocabulary use, and appropriateness of expressions. Grammatical accuracy evaluates the correctness of sentence structures. Appropriateness of vocabulary use measures the suitability and variety of vocabulary in context. Appropriateness of expressions assesses the naturalness and diversity of expressions, as well as the appropriateness of sentence length.

The second category, *organization*, contains four subcategories: appropriateness of internal structure, organization of essay structure, coherence of paragraphs, and appropriateness of length. Appropriateness of internal structure measures how well thesis and supporting sentences are connected. Organization of essay structure evaluates the presence and proportionality of introduction, body, and conclusion. Coherence of paragraphs considers the use of transitions and discourse markers. Appropriateness of length checks adherence to the required word count.

The third category, *content*, includes clarity of topic, detailed explanation, and creativity of thought. Clarity of topic measures whether the topic is clearly presented and consistently developed. Detailed explanation evaluates the depth and variety of elaboration. Creativity of thought captures originality, logical reasoning, and the inclusion of new perspectives.

Table 1 presents the rubric categories, subcategories, and their descriptions.

To ensure reliable application of this rubric, we implemented a multi-step annotator training and calibration process. Two annotators, each holding an M.A. in Korean Language Education and with extensive teaching and rating experience, participated in the scoring. Prior to rating, they underwent two training sessions led by the researcher. In the first session, the raters conducted a content and response-process check by independently scoring 15 sample L2 essays while verbalizing their reasoning. This procedure allowed the identification and clarification of ambiguous descriptors. Following this step, 20% of the essays were independently scored. Afterward, the raters engaged in a one-hour focus group discussion to review discrepancies, evaluate item fit, assess the rubric's usability for L2 writing, and identify potential missing descriptors. The raters unanimously agreed that the rubric adequately cap-

Table 1 Rubric categories and subcategories used for evaluating Korean L2 essays

Category	Subcategories	Descriptions
Lexis and grammar	Grammatical accuracy	Correct use of sentence structures
	Appropriateness of vocabulary use	Vocabulary suitability and variety
	Appropriateness of expressions	Variety and naturalness of expressions; sentence length appropriateness
Organization	Appropriateness of internal structure	Coherence between thesis and supporting sentences
	Organization of essay structure	Presence and proportionality of introduction, body, and conclusion
	Coherency of paragraphs	Logical flow and use of discourse markers
	Appropriateness of length	Adherence to required word count
Content	Clarity of topic	Clear presentation of the topic and relevance of content
	Detailed explanation	Depth and variety in explanation
	Creativity of thought	Originality, logical structure, and new perspectives

Table 2 Cohen's kappa scores for inter-annotator agreement across learner subgroups–foreign beginners (FB), foreign intermediates (FI), heritage beginners (HB), and heritage intermediates (HI)–along with the overall score (All)

FB	FI	HB	HI	All
0.8253	0.8752	0.7971	0.7721	0.8401

tured L2 learner performance, confirming its validity for this context. Once consensus was reached, the two raters independently scored all essays, and inter-rater reliability was measured.

This process reflects best practices in standardized assessment, where inter-rater reliability, operationalized rubrics, and calibration are critical for consistency. Four points were assigned to each of the ten criteria, scored on a scale from 0 to 3, yielding a maximum possible score of 30 points per essay.

Cohen's kappa was used to assess inter-rater reliability. Table 2 reports scores for each learner subgroup–foreign beginners (FB), foreign intermediates (FI), heritage beginners (HB), and heritage intermediates (HI)–as well as the overall score. Results indicate consistently high agreement, with the strongest reliability for foreign intermediates ($\kappa = 0.8752$) and the weakest for heritage intermediates ($\kappa = 0.7721$). The overall score ($\kappa = 0.8401$) demonstrates robust reliability.

To interpret the subgroup differences, we examined the distribution of error types contributing to kappa variability. While overall patterns align with those reported in Table 3, the HI group exhibits a higher proportion of spelling errors, reflecting greater lexical diversity, whereas functional morpheme–related errors occur less frequently. This distribution helps explain the comparatively lower kappa for heritage intermedi-

Table 3 Top five grammatical error types and their relative distributions across learner subgroups: foreign beginners (FB), foreign intermediates (FI), heritage beginners (HB), and heritage intermediates (HI)

FB		FI		HB		HI	
0.3729	R:SPELL	0.3975	R:SPELL	0.4620	R:SPELL	0.4818	R:SPELL
0.0536	R:NOUN+ADP → NOUN+ADP	0.0567	R:NOUN → NOUN+ADP	0.0457	R:NOUN → NOUN+ADP	0.0411	R:NOUN → NOUN+ADP
0.0421	R:ADV → ADV	0.0405	R:NOUN+ADP → NOUN+ADP	0.0249	R:NOUN+ADP → NOUN+ADP	0.0332	R:NOUN+ADP → NOUN+ADP
0.0370	R:NOUN → NOUN+ADP	0.0344	R:VERB+PART → VERB+PART	0.0218	R:VERB+PART → VERB+PART	0.0268	R:VERB+PART → VERB+PART
0.0140	R:VERB+PART → VERB+PART	0.0304	R:ADV → ADV	0.0202	R:ADV → ADV	0.0174	R:ADV → ADV

ates: annotators tend to agree on functional morpheme errors, but greater variability in lexical choice increases the likelihood of divergence in scoring.

In sum, inter-rater agreement ranged from substantial to almost perfect ($\kappa = 0.77$ –0.88), with an overall score of $\kappa = 0.84$. Mean scores were closely aligned, though one annotator tended to assign slightly higher values across categories, with differences remaining marginal (< 0.05). These results, together with the diagnostic error distribution, indicate that the rubric-based scoring scheme is both reliable and sensitive to genuine areas of linguistic variability.

On this basis, by adopting a detailed, pre-established rubric, conducting structured annotator training, and validating inter-rater reliability, KoLLA v2.0 establishes a robust benchmark for evaluating Korean L2 essays. This enriched dataset supports not only pedagogically meaningful assessment but also the development of automated essay scoring systems grounded in standardized evaluation criteria.

These results confirm that our rubric-based scoring procedure achieved robust reliability, while also remaining sensitive to genuine linguistic variability. By combining a validated rubric, structured annotator training, and systematic reliability checks, KoLLA v2.0 establishes the first rubric-scored Korean learner corpus. This resource provides a standardized benchmark for evaluating Korean L2 writing proficiency and for developing automated essay scoring systems grounded in pedagogically meaningful criteria.

4 Findings and discussion

The augmented KoLLA corpus provides new insights into both the linguistic features of Korean L2 learner writing and the methodological value of multi-layered annotation. In this section, we discuss our findings in two parts: (i) the implications of multi-reference GEC for capturing correction variability and evaluating system performance, and (ii) the significance of rubric-based scoring for assessing overall writing quality and learner proficiency in a standardized and pedagogically meaningful way.

4.1 Multi-reference grammatical error annotation

The enhancements to the KoLLA corpus provide insights into grammatical error patterns in Korean L2 writing, with particular emphasis on the benefits of multi-reference GEC. Table 3 summarizes the distribution of grammatical error types in the multi-reference KoLLA corpus. In our annotation scheme, particle omission is explicitly represented as `NOUN` $\rightarrow$ `NOUN+ADP`, directly marking the absence of a postposition. By contrast, when only the lexical morpheme is replaced while functional morphemes such as postpositions or verbal endings remain correct, the error is classified as a spelling error. Thus, spelling errors in our framework capture substitutions of lexical morphemes (e.g., noun stems, verbal roots) without altering morpho-syntactic markers. Under this scheme, spelling errors emerge as the dominant error type across proficiency levels, paralleling findings in Chinese learner corpora (Gu et al., 2025), while punctuation errors–frequent in English corpora such as W&I (Zeng

et al., 2024)–are largely absent in Korean. Alongside these trends, particle-related errors (NOUN →NOUN+ADP) remain among the most frequent categories, reflecting the well-documented difficulty learners face with postpositions in agglutinative languages (Lee et al., 2009, 2012; Israel et al., 2013). Multi-reference annotations therefore capture both cross-linguistic similarities and Korean-specific challenges, particularly the joint prominence of spelling variability and postposition omission.

These patterns point to the usefulness of a multi-reference design for representing genuine variability in Korean learner writing. To the best of our knowledge, the multi-reference KoLLA corpus is the first resource of its kind specifically tailored for Korean L2, addressing inherent linguistic variability and enabling more comprehensive and contextually appropriate evaluations.

Single-reference GEC datasets may overlook natural variability in learner language, potentially penalizing alternative yet valid corrections. In contrast, multi-reference annotations capture the diversity of valid grammatical structures–especially critical given Korean's syntactic and morphological flexibility. Case markers, verb endings, and compound verbs often permit multiple correct realizations, each reflecting a valid linguistic choice depending on discourse context.

To quantify the relationship between the original and the newly added corrections, we conducted an agreement analysis using the M2 evaluation framework. Treating the original annotations as the reference and our new annotations as the hypothesis, we obtained an F1 score of 0.3881, with 708 true positives, 1173 false positives, and 1060 false negatives. This level of divergence highlights the limits of single-reference evaluation and mirrors results reported for English and Chinese multi-reference corpora. Such moderate overlap reflects the considerable variability that naturally arises in grammatical error correction, where multiple valid corrections coexist for a single learner sentence.[2] In Korean–a morphologically rich and agglutinative language–this variability is particularly pronounced, as alternative corrections frequently diverge in case marking, verb endings, or lexical choice while still preserving grammaticality and meaning. These findings reinforce the necessity of multi-reference annotation, which more faithfully represents the range of acceptable corrections in learner writing and enables fairer evaluation of automated GEC systems.

In addition to morphosyntactic variability, lexical flexibility further contributes to correction diversity, as synonymous substitutions frequently preserve meaning while differing in surface form. This characteristic underscores the importance of evaluation frameworks that prioritize semantic adequacy over rigid form matching.

Future analyses will expand on these observations by conducting a quantitative error breakdown, examining correlations with proficiency levels, and assessing how variation in correction strategies influences automated model performance.

4.2 Rubric-based scoring for writing assessment

Beyond grammatical error correction, the introduction of rubric-based scoring provides a broader framework for evaluating Korean L2 writing proficiency. Whereas GEC focuses on grammatical accuracy at the sentence level, rubric-based scoring

[2] A detailed comparison of multi-reference GEC evaluation is provided in Appendix B.

assesses higher-order dimensions of writing, including coherence, structure, and content development. In this work, we foreground rubric-based scoring as a systematic means of evaluating these discourse-level features, which are often overlooked by surface-oriented GEC approaches. By incorporating scores aligned with the Korean National Language Institute guidelines, the KoLLA corpus establishes a standardized protocol for evaluating L2 writing performance.

Analysis of the rubric scores reveals a strong correlation between grammatical accuracy and overall writing proficiency. Essays with fewer grammatical errors consistently received higher scores, underscoring the interdependence of grammatical competence and discourse-level writing skills. Lexical diversity also emerges as a distinguishing factor across proficiency levels: essays with broader vocabulary ranges tend to achieve higher ratings, highlighting the role of lexical variation in proficiency assessment.

Structural complexity constitutes another key indicator of writing ability. Higher-scoring essays display stronger paragraph organization, more effective use of discourse markers, and smoother transitions between ideas. Nonetheless, rubric scores also reveal persistent challenges among intermediate learners, including underdeveloped arguments, illogical sequencing, and inappropriate rhetorical structures. These findings suggest that instructional interventions targeting discourse-level organization are essential even after learners achieve greater grammatical control.

Differences also emerge across essay types. Argumentative essays often contain more discourse-related errors, particularly in structuring logical arguments and supporting claims with evidence. In contrast, personal essays tend to score higher in fluency and coherence but may exhibit lexical repetition. Such distinctions indicate that instructional approaches should be tailored to the demands of each genre.

Across the rubric dimensions, the findings show how rubric-based scoring distinguishes broad trends across both proficiency levels and genres, and this pattern is also visible at the individual level. To illustrate this at the individual level, we include a brief example from the dataset. An advanced learner's essay on environmental policy received full scores in organization and content development (3 out of 3 each), but slightly lower in lexis and grammar (2 out of 3), reflecting sophisticated argumentation with occasional lexical and morphological errors. In contrast, a lower-intermediate learner's essay on the same topic scored 1 out of 3 across categories, with shorter, less cohesive paragraphs and frequent grammatical errors. These examples highlight the rubric's ability to capture qualitative distinctions in L2 writing beyond sentence-level accuracy, demonstrating its value for both pedagogical feedback and research on Korean learner language.

4.3 Implications and future directions

The integration of multi-reference GEC and rubric-based scoring in the KoLLA corpus carries important implications for both Korean L2 learning and the development of automated NLP systems. For language education, the dataset provides fine-grained insights into common learner errors, allowing instructors to design targeted curricula that address persistent challenges such as postposition omission, honorific mismatches, and discourse-structural difficulties. Rubric-based scoring further extends

these applications by enabling teachers to deliver feedback on higher-order writing skills–such as discourse coherence, lexical variety, and content development–alongside grammatical accuracy, thereby supporting more balanced and pedagogically relevant instruction. The refined annotation framework thus serves as a valuable foundation for the development of instructional materials aimed at strengthening both grammatical competence and overall writing fluency.

For computational modeling, the inclusion of multi-reference annotations reduces the penalization of alternative but grammatically acceptable corrections, thereby facilitating the training of flexible and context-aware GEC systems. By incorporating diverse correction strategies, models can better reflect the inherent linguistic variability of Korean learner writing. Moreover, the multi-reference design provides a robust benchmark for evaluating GEC systems, ensuring that model outputs are judged against multiple valid references rather than a single, rigid gold standard.

In the context of AES, rubric-based scores establish a foundational dataset for training AI-driven assessment models that align closely with human grading practices. By jointly modeling grammatical, lexical, and discourse-level features, these systems can deliver more nuanced feedback to learners, supporting both formative feedback and summative evaluation. The integration of rubric scoring into neural architectures opens promising directions for developing personalized and pedagogically meaningful feedback systems.

Future research should explore scaling multi-reference GEC by incorporating additional annotators and expanding the range of linguistic variation represented. Further extensions could involve annotating discourse-level errors, which would broaden the dataset's utility for both GEC and AES research. Additionally, the integration of rubric-based scoring into neural assessment models offers opportunities to develop more sophisticated feedback systems for Korean L2 writing instruction.

Across the two analyses, the results point to the need for using both annotation layers. These findings underscore the necessity of integrating multi-reference grammatical error annotation and rubric-based scoring in Korean learner corpora. Such integration offers a more comprehensive framework for analyzing L2 writing, bridging the gap between error correction, writing assessment, and instructional design.

4.4 Summary of findings

The enhanced KoLLA corpus (KoLLA v2.0) demonstrates the benefits of integrating multi-reference grammatical error annotations with rubric-based scoring to provide a more holistic evaluation of Korean L2 writing. Multi-reference GEC captures the linguistic variability and ambiguity intrinsic to Korean, while rubric-based scoring offers a structured, pedagogically grounded framework for assessing broader dimensions of writing proficiency. In combination, these two components support both computational development and classroom practice, paving the way for more accurate, equitable, and informative evaluation of learner language.

5 Conclusion

This study enriches the KoLLA Korean learner corpus by incorporating multi-reference GEC annotations and rubric-based scoring, addressing critical gaps in Korean L2 writing research. By transforming KoLLA into a multi-reference GEC resource, we capture the inherent variability of grammatical correction, while rubric-based scoring establishes a standardized approach for assessing learner proficiency. These two enhancements jointly position KoLLA v2.0 as a linguistically principled and computationally robust foundation for future work.

Rather than reiterating specific findings, we emphasize KoLLA v2.0 as both a benchmark and a roadmap. As a benchmark, it provides a reliable resource for evaluating GEC and AES in Korean, supporting fairer comparisons across systems and approaches. As a roadmap, it points toward future expansion: larger-scale data collection across genres and proficiency levels, discourse-level and pragmatic annotation, and the integration of neural AES models for more adaptive and pedagogically relevant feedback. In addition, Korean's agglutinative morphology poses unique challenges for GEC, including complex error spans involving bound morphemes, ambiguity in word segmentation, and difficulty in defining precise correction boundaries. These linguistic characteristics require annotation decisions that balance granularity with cross-linguistic consistency. We acknowledge that the current corpus size is relatively modest due to the manual, multi-step nature of annotation, but we view this deliberate methodology as essential for ensuring data quality and reliability. Future work will focus on expanding the corpus to cover a wider range of proficiency levels and writing genres–including argumentative, expository, and narrative texts–in order to strengthen the generalizability of our findings.

By addressing both linguistic and computational challenges, this study contributes to the continued development of Korean learner corpora and reinforces the importance of high-quality, systematically annotated data for L2 writing research. The enriched KoLLA v2.0 thus serves as both a benchmark for immediate evaluation and a roadmap for future expansion, paving the way for more sophisticated tools that support Korean L2 learners and advance cross-linguistic research on second-language writing.

Appendix A: Dataset description

The enhanced KoLLA dataset (KoLLA v2.0) comprises two primary components designed to support GEC and AES:

1. **GEC** `M2` **File**: This file follows the standardized `M2` format widely used in grammatical error annotation tasks. It contains pairwise alignments between original learner sentences and their corrected forms, annotated with detailed error tags based on the extended MRU typology adapted for Korean. Each sentence is paired with two reference corrections, enabling multi-reference evaluation. Annotations specify the location, type, and nature of each error, capturing both surface-level and structural changes.

2. **AES .csvFiles for FB, FI, HB, and HI**: These files contain rubric-based scoring data for four learner groups–foreign beginners (FB), foreign intermediates (FI), heritage beginners (HB), and heritage intermediates (HI). Each CSV file includes 250 columns representing 10 rubric dimensions (Table 1) across 25 essays. Scores are based on the Korean National Language Institute's writing evaluation criteria, covering expression, structure, and content. Each file provides two rows corresponding to independent ratings by two trained annotators, allowing for inter-rater reliability analysis and fine-grained evaluation of writing performance across demographic and proficiency subgroups.

Appendix B: Multi-reference GEC evaluation

To illustrate the impact of multi-reference annotation on GEC evaluation, we conducted pilot experiments using a KoBART pretrained model (Yoon et al., 2023). The model was tasked with generating corrections for learner sentences, and performance was assessed under two conditions: (i) using only the original GEC annotations as a single reference, and (ii) using both the original and the newly introduced corrections as multiple references.

Evaluation followed the standard M2 framework, with precision, recall, and $F_{0.5}$ scores computed. When multiple references are available for a sentence, the system output is evaluated against each reference independently, and the reference yielding the highest *F*-score is selected. This sentence-level best-match strategy ensures that systems receive credit for any valid correction, even if it matches only one of several plausible human annotations. This design accommodates the variability and ambiguity of grammatical error correction, where multiple corrections may be equally acceptable.

Table 4 Evaluation results of the KoBART-based GEC system under single- and multi-reference settings

Evaluation setting	TP	FP	FN	Precision	Recall	$F_{0.5}$
Single reference (original only)	353	834	1415	0.2974	0.1997	0.2709
Multi-reference (original + proposed)	495	692	1029	0.4170	0.3248	0.3946

Table 4 presents the results. As expected, evaluation with multiple references yields consistently higher recall and overall $F_{0.5}$, since valid alternative corrections are less likely to be penalized. This demonstrates the value of employing a multi-reference GEC dataset, particularly for Korean, where syntactic flexibility and lexical variation frequently allow more than one grammatically acceptable correction.

These findings confirm that incorporating multiple references provides a more faithful evaluation of system performance by recognizing the diversity of valid corrections in Korean learner writing. This approach reduces the risk of underestimating performance due to rigid single-reference matching and aligns evaluation more closely with pedagogical and linguistic realities.

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Author contributions J.S. supervised the development of the AES scoring dataset, and K.L. supervised the development of the multi-reference GEC dataset. J.P. oversaw the entire process. All authors contributed to the manuscript and approved the final version.

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Data availability The dataset is available at https://zenodo.org/records/15287128 (https://doi.org/10.5281/zenodo.15287128) and is released under the GNU General Public License Version 3, consistent with the licensing of the original resource.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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